

Synent Technologies

Data Science Internship

Task 1 – Titanic Data Cleaning & Preprocessing

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Internship Task Report

1. Objective

The objective of this task was to clean and preprocess the Titanic dataset in order to produce a complete, consistent, and properly formatted version suitable for further data analysis or machine learning work. This included handling missing values, verifying data types, checking for duplicate records, and standardizing column names.

2. Dataset Overview

The dataset used for this task was **Titanic-Dataset.csv**, containing 891 rows and 12 columns. The original columns were: PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, and Embarked.

3. Initial Data Quality Assessment

The dataset was inspected for column names, data types, missing values, and duplicate rows prior to cleaning. The number of missing values found in each column is summarized below.

Column	Missing Values
PassengerId	0

Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

The dataset was also checked for duplicate rows; none were found (0 duplicate rows).

4. Data Cleaning and Preprocessing

Based on the initial assessment, the following preprocessing steps were carried out:

- **Missing Age values:** Replaced using the median age of the dataset.
- **Missing Embarked values:** Replaced using the mode (most frequent value) of the column.
- **Missing Cabin values:** Replaced with the placeholder value "Unknown", since the majority of entries in this column were missing.
- **Duplicate rows:** The dataset was checked using `drop_duplicates()`. Since the dataset originally contained 0 duplicate rows, no records were removed at this step.
- **Data type conversion:** Relevant columns were converted to appropriate data types to ensure consistency.
- **Column renaming:** All columns were renamed to clear, descriptive snake_case names for readability and ease of use in later analysis.

The final column names after renaming are listed below:

passenger_id, survived, passenger_class, name, sex, age, siblings_spouses, parents_children, ticket, fare, cabin, embarked

5. Final Results

After the cleaning and preprocessing steps were applied, the dataset was verified to confirm that all data quality issues had been resolved.

Metric	Result
Final dataset size	891 rows x 12 columns
Final missing values	0
Final duplicate rows	0
Output file	titanic_cleaned.csv

6. Tools Used

Python, Pandas, and Google Colab / Jupyter Notebook were used to complete this task.

7. Conclusion

Through this task, the Titanic dataset was successfully cleaned and preprocessed. Missing values in the Age, Embarked, and Cabin columns were addressed, data types were verified, column names were standardized to snake_case, and the dataset was confirmed to be free of duplicate records. The resulting dataset, saved as titanic_cleaned.csv, is now complete and consistent, and is suitable for use in further data analysis or machine learning tasks.